

HYPERAUTOMATION LABS

THE OPEN-SOURCE HEDGE FUND

Multi-Agent Trading Architecture
That Mirrors Wall Street

54K Stars • Free • Open Source • Run It On Your Laptop

54K

GITHUB
STARS

FREE

OPEN
SOURCE

6

AI AGENTS

6+

LLM
PROVIDERS

by Hyperautomation Labs

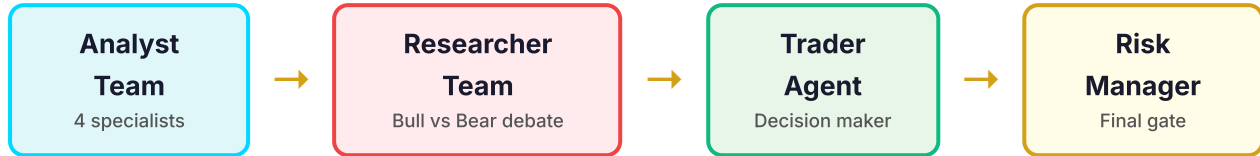
GitHub: [TauricResearch/TradingAgents](https://github.com/TauricResearch/TradingAgents)

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The Multi-Agent Trading System

How 6 AI agents replicate the decision-making process of a professional hedge fund.

The Org Chart



The 4 Analysts

Market Analyst

Tracks price action, volume patterns, moving averages, RSI, MACD, and Bollinger Bands. Identifies technical entry/exit signals and trend direction.

News Analyst

Scans Reuters, Bloomberg, SEC filings, and earnings calls for material events. Identifies catalysts, regulatory risks, and sector-wide impacts.

Social Media Analyst

Monitors Reddit, X/Twitter, and StockTwits for retail sentiment. Tracks mention velocity, sentiment shifts, and viral momentum before price moves.

Fundamentals Analyst

Deep-dives balance sheets, P/E ratios, revenue growth, margins, and competitive positioning. Compares against sector benchmarks and historical averages.

The Debate Mechanism

HOW THE DEBATE WORKS

Two researchers argue opposing positions:

- **Bull Researcher** — builds the strongest possible case for buying. Uses all 4 analyst reports to find upside catalysts, positive trends, and growth signals.
- **Bear Researcher** — builds the strongest case against. Finds downside risks, overvaluation signals, negative sentiment, and red flags.
- **Research Judge** — evaluates both arguments impartially. Identifies which side made stronger evidence-based claims. Produces a final synthesis.

The result: No blind spots. Every decision has been stress-tested from both sides before the Trader Agent ever sees it.

Decision Flow

THE PIPELINE

4 Analyst Reports → **Bull/Bear Debate** → **Trader Decision** (timing, position size, entry point) → **Risk Manager Gate** (volatility, liquidity, portfolio balance checks) → **APPROVED or REJECTED**

The Trader cannot override the Risk Manager. Checks and balances are enforced by the architecture, not by human discipline.

How to Run It

From zero to running your own multi-agent trading system in under 10 minutes.

Installation

```
# Clone the repository
git clone https://github.com/TauricResearch/TradingAgents.git
cd TradingAgents # Install dependencies
pip install . # Run it
tradingagents
```

Python API Example

```
from tradingagents import TradingAgents # Initialize with your preferred LLM
ta = TradingAgents( llm_provider="anthropic", # or "openai", "google", etc.
                    model="claude-sonnet-4-20250514", ) # Run analysis on any ticker
result = ta.run( ticker="NVDA", period="2026-04", ) # Get the decision
print(f"Action: {result.action}") # BUY / SELL / HOLD
print(f"Confidence: {result.confidence}")
print(f"Risk: {result.risk_assessment}")
```

Supported LLMs

PROVIDER	MODELS	LOCAL?	NOTES
OpenAI	GPT-4o, GPT-4.1, o3	No	Best out-of-box experience
Anthropic	Claude Sonnet 4, Opus 4	No	Strong reasoning for debate
Google	Gemini 2.5 Pro/Flash	No	Good price/performance ratio
xAI	Grok	No	Real-time data integration
DeepSeek	DeepSeek-V3, R1	No	Cost-effective alternative
Ollama	Any local model	Yes	Fully private, no API keys needed

Docker Setup (Local Models)

```
# Pull and run Ollama
docker run -d --name ollama -p 11434:11434 ollama/ollama
docker exec ollama ollama pull llama3.1:70b # Configure TradingAgents to use local
ta = TradingAgents( llm_provider="ollama", model="llama3.1:70b", )
```

Key Features

DECISION LOGGING
Every agent decision is logged with full reasoning chains. Review why each analyst flagged specific signals, how the debate unfolded, and what the Trader prioritized. Learns from past decisions.

CHECKPOINT RECOVERY
Built on LangGraph's state management. If the system crashes mid-analysis, it resumes from the last checkpoint — no wasted API calls, no lost work. Production-grade reliability.

The Pattern Applies to Everything

The hedge fund is a template. The multi-agent debate is the innovation.

Adapt the Architecture

 **Legal**

Replace analysts with legal researchers. Bull/Bear becomes prosecution vs defense. The Trader becomes a strategy recommender. Risk Manager checks precedent and compliance.

 **Business Strategy**

Analysts cover market, competitive, financial, and operations data. Researchers debate optimistic vs pessimistic scenarios. Trader recommends strategy. Risk Manager stress-tests assumptions.

 **Medical**

Analysts become specialist consultants (cardiology, radiology, pathology). Researchers debate treatment options. Trader recommends. Risk Manager checks contraindications and patient history.

 **Research**

Analysts are domain-specific literature reviewers. Researchers debate methodology and conclusions. Trader synthesizes findings. Risk Manager checks for bias and reproducibility.

How to Adapt It

THE RECIPE

- Fork the repo** — github.com/TauricResearch/TradingAgents
- Replace the analysts** — Swap market/social/news/fundamentals with your domain experts
- Keep the debate** — The bull/bear adversarial pattern is the core innovation. Adapt the framing but keep the structure
- Customize the risk checks** — Replace volatility/liquidity/portfolio with your domain's safety gates
- Choose your LLM** — Use Anthropic for strong reasoning, Ollama for privacy, GPT-4o for speed

Resources

Links

- **GitHub:** github.com/TauricResearch/TradingAgents
- **arXiv Paper:** arxiv.org/abs/2412.20138
- **Built on:** LangGraph (state management + agent orchestration)

Why This Matters

- Hedge funds charge 2/20 for this decision-making process
- This repo gives it away for free
- The debate architecture eliminates confirmation bias
- Runs on your laptop with local models

Quick Reference

AGENT	ROLE
Market Analyst	Technical signals
Social Analyst	Retail sentiment
News Analyst	Catalysts & risks
Fundamentals	Financial health
Bull Researcher	Upside case
Bear Researcher	Downside case
Trader	Final decision
Risk Manager	Safety gate

Your Guides — Build & Ship Faster

- Claude Code Guide** — hyperautomationlabs.gumroad.com/l/claude-code-guide
- Codex Guide** — hyperautomationlabs.gumroad.com/l/codex-guide

3. **Claude Cowork Sales Guide** — hyperautomationlabs.gumroad.com/l/claude-cowork-sales

4. **Claude Certified Architect Prep** — hyperautomationlabs.gumroad.com/l/claude-certified-architect-prep

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